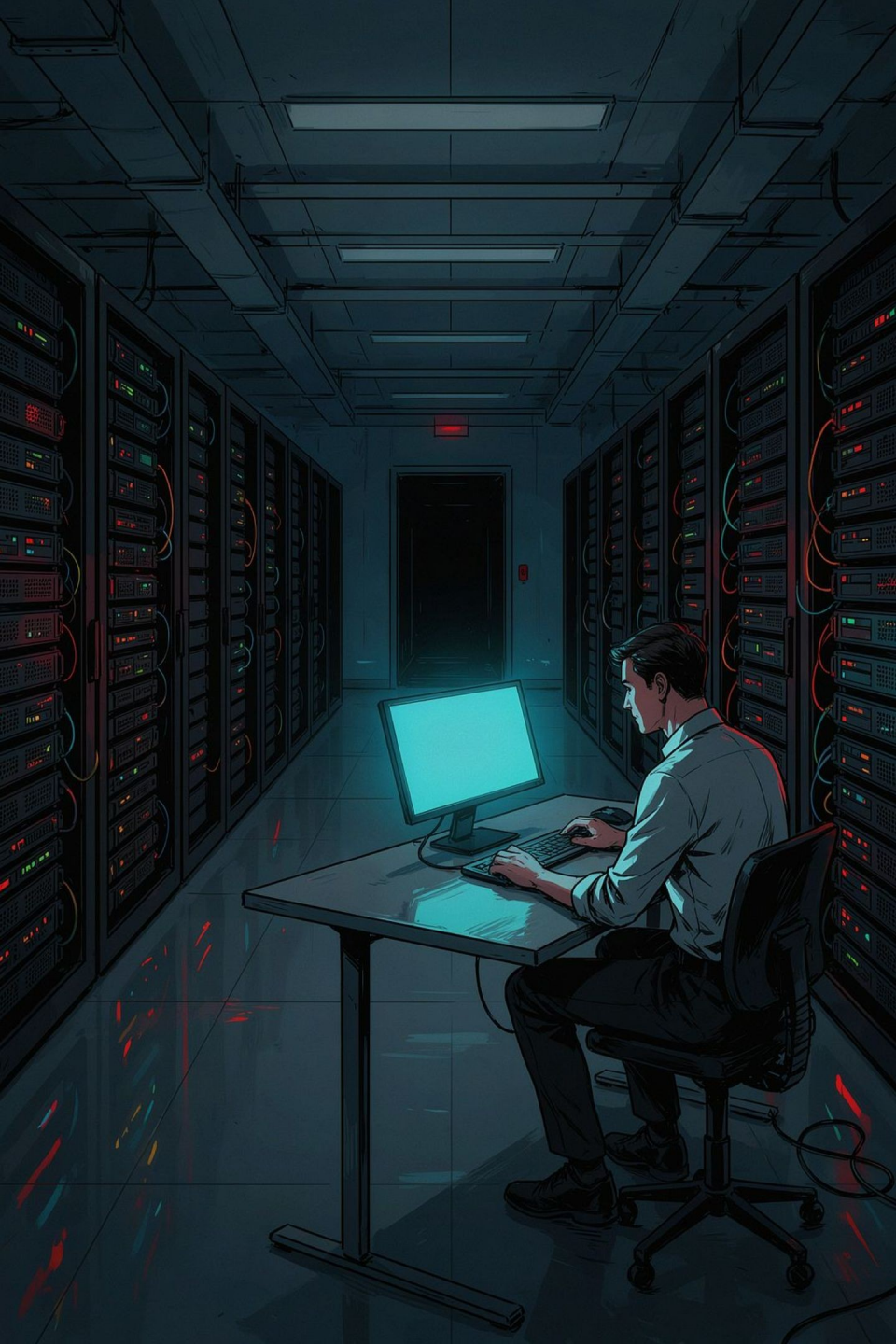




Jenkins Administration

Managing and Maintaining Enterprise Jenkins Servers

Audience: DevOps Engineers, Build Engineers, CI/CD Administrators

Jenkins Administration

Topics Covered

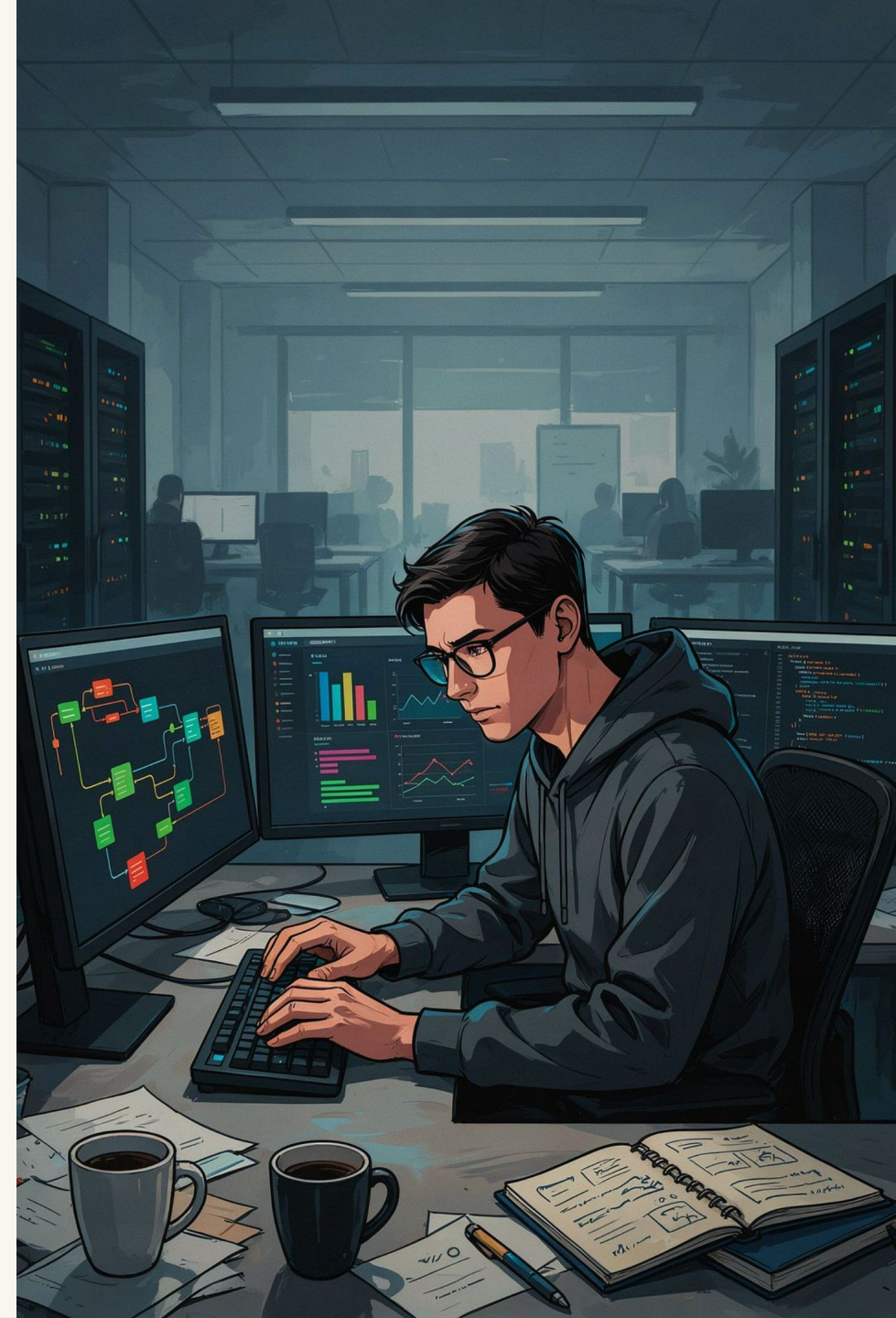
Topics Covered

- Jenkins Home Directory
- Global Configuration
- System Configuration
- Tool Configuration
- Plugin Management
- Best Practices
- Hands-on Demonstration

Learning Outcomes

By the end of this session you will be able to:

- Understand Jenkins architecture
- Configure Jenkins globally
- Install and manage plugins
- Configure build tools
- Manage administrator settings
- Maintain a production Jenkins server



Responsibilities of a Jenkins Administrator

A Jenkins administrator is responsible for maintaining a healthy Jenkins environment.

Primary responsibilities include:

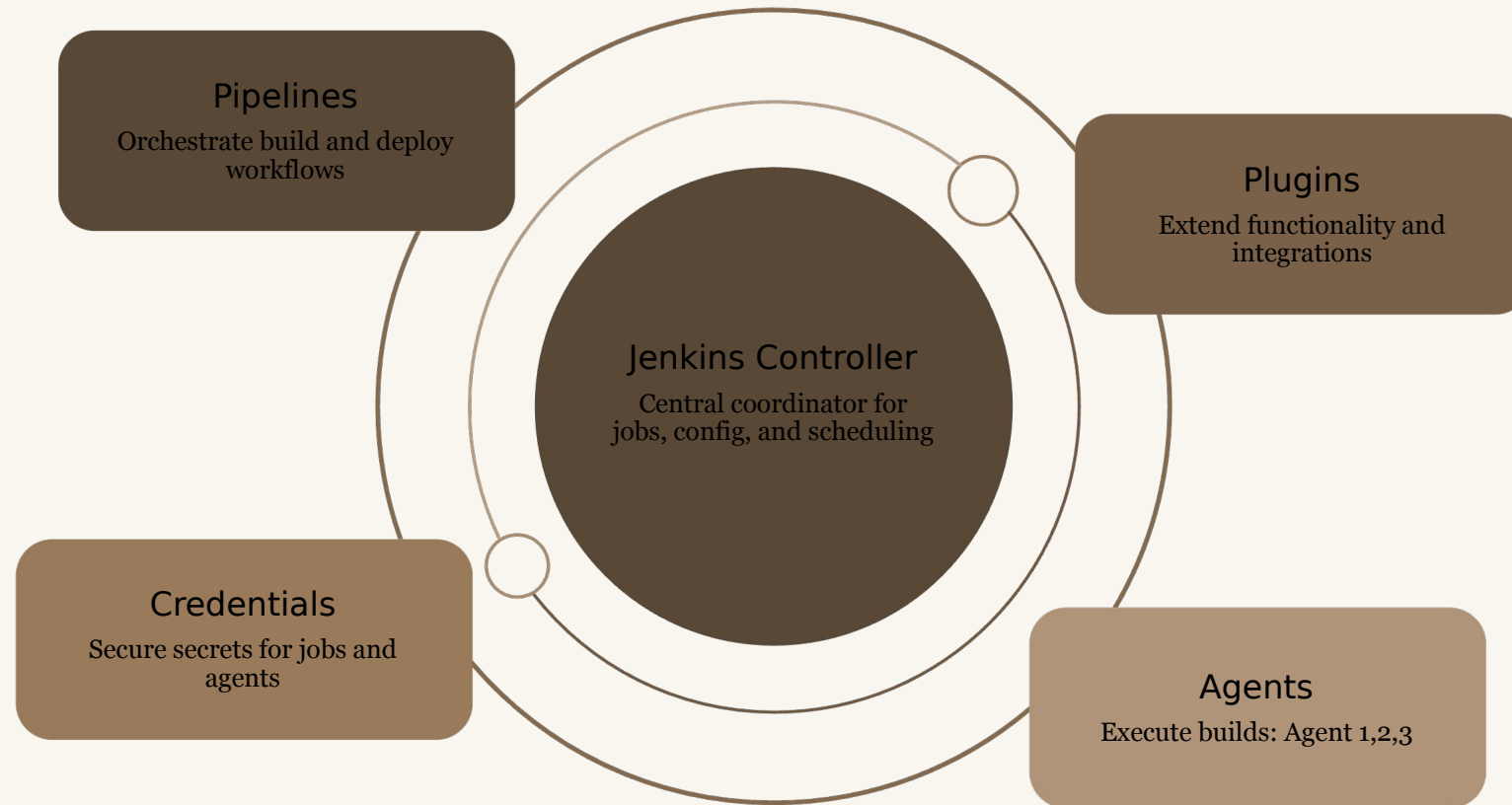
- Installing Jenkins
- Configuring system settings
- Managing users
- Installing plugins
- Configuring build tools
- Managing credentials
- Configuring agents
- Backup and recovery
- Security hardening
- Monitoring server health

Administrator vs Developer

Administrator	Developer
Installs Jenkins	Creates Pipelines
Configures Tools	Writes Jenkinsfile
Installs Plugins	Builds Applications
User Management	Uses Build Jobs
Security	Deployment Logic

Jenkins Architecture Review

Jenkins Components



Controller

- Scheduling
- UI
- Plugin execution

Agents

- Build execution
- Testing
- Deployment

Storage

- Job configuration
- Workspace
- Build history



What is JENKINS_HOME?

The Jenkins Home directory contains everything Jenkins needs.

Typical location

Linux

`/var/lib/jenkins`

Docker

`/var/jenkins_home`

Windows

`C:\ProgramData\Jenkins`

Jenkins Home Directory Structure

JENKINS_HOME

- ├─ config.xml
- ├─ credentials.xml
- ├─ users/
- ├─ jobs/
- ├─ workspace/
- ├─ plugins/
- ├─ nodes/
- ├─ secrets/
- ├─ updates/
- ├─ logs/
- └─ fingerprints/

Important Directories

Jobs

Stores every Jenkins job

Workspace

Stores source code during builds

Plugins

Installed plugins

Secrets

Encryption keys

Users

User profiles

Nodes

Agent configuration

Exploring Jenkins Home

Important Files

config.xml

Main Jenkins configuration

credentials.xml

Encrypted credentials

plugins/

Installed plugins

users/

User configuration

jobs/

Every job has its own folder

Example

```
jobs/  
  MyApp/  
    config.xml  
  builds/  
  workspace/
```

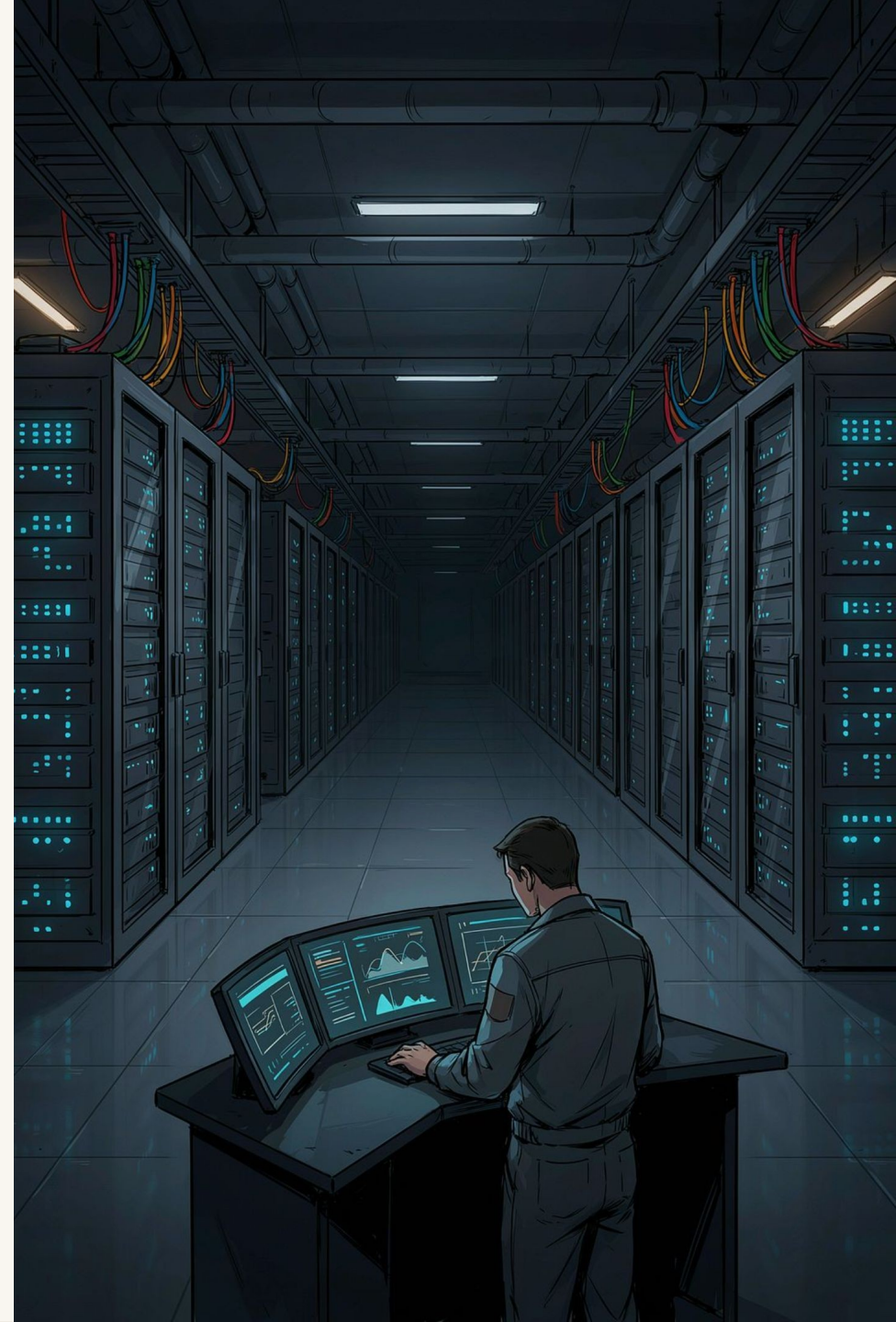
Best Practices for Jenkins Home

Do

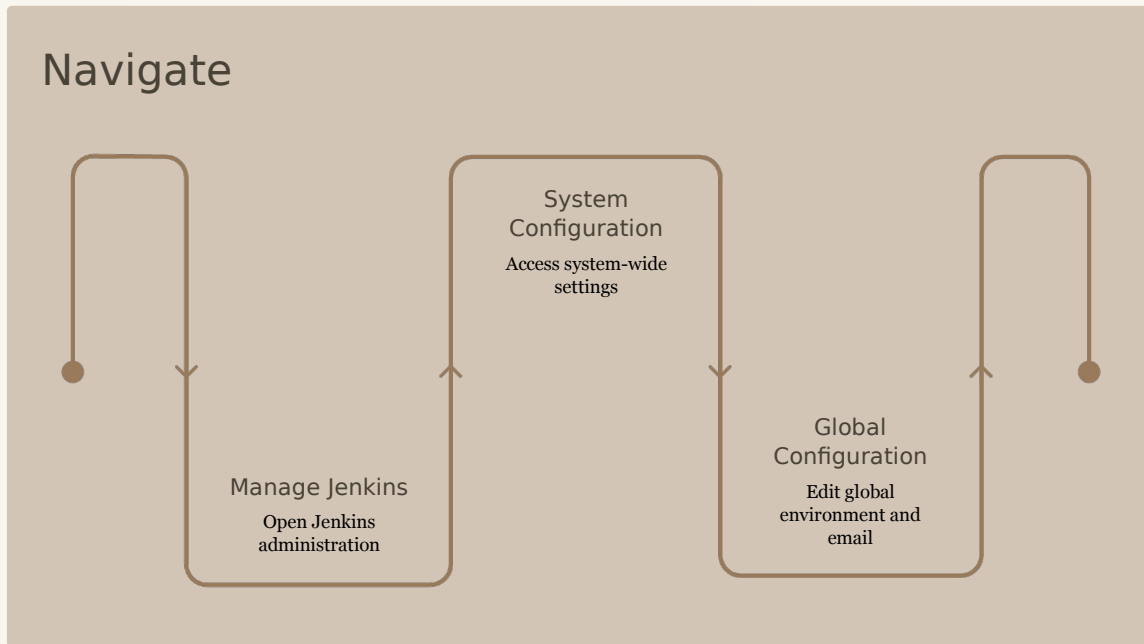
- Take daily backups
- Monitor disk usage
- Store on dedicated storage
- Restrict permissions
- Monitor workspace size

Don't

- Edit XML while Jenkins is running
- Delete plugin folders manually
- Delete workspaces randomly
- Share JENKINS_HOME across multiple controllers



Jenkins Global Configuration Overview



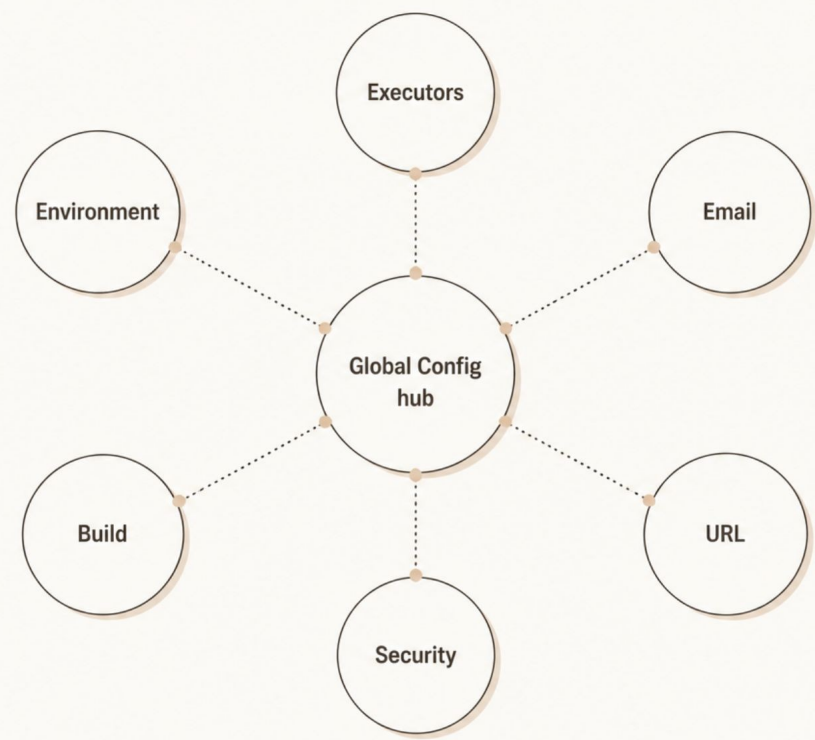
Global Configuration controls

- Global settings
- Environment variables
- Email
- Security
- Build behavior

Global Configuration Sections

Major sections include

Jenkins URL	Number of Executors
Environment Variables	SMTP
Global Properties	Quiet Period
Build Timeout	SCM Checkout Retry Count





Configure Jenkins URL

Example

`http://jenkins.company.com`

Best Practice

Always use DNS instead of IP address

Purpose

- Email links
- Webhook callbacks
- Build notifications
- External integrations

Configure Environment Variables

Example

- JAVA_HOME
- MAVEN_HOME
- DOCKER_HOST
- AWS_REGION
- ENVIRONMENT

These variables become available to every Pipeline.

Example

`ENV=Production`

Used across all jobs.

Number of Executors

Executors determine how many jobs Jenkins can execute simultaneously.

Example

Server	16 CPU
Suggested executors	12

 **Best Practice:** Avoid configuring executors equal to total CPU cores.

Executor Layout

Executor 1

Executor 2

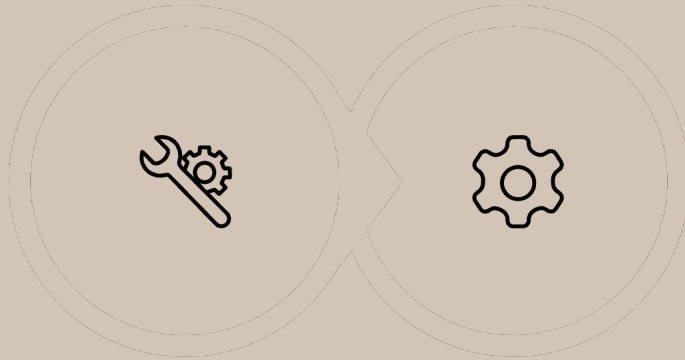
Executor 3

Executor 4

... Executor 12

System Configuration

Navigate



Manage
Jenkins

System

This page configures

- Jenkins URL
- SMTP
- Slack
- GitHub
- Global Libraries
- Environment Variables
- Build Discarders

SMTP Configuration

Purpose

Send

- Build notifications
- Failure alerts
- User notifications

Example

Setting	Value
SMTP Server	smtp.office365.com
Port	587
Authentication	Username / Password
Use TLS	Yes

Global Pipeline Libraries

Purpose

Reuse common pipeline code

Instead of

Jenkinsfile 1000 lines

Use

```
@Library('shared-library')
```

Benefits

Reusable

Centralized

Version controlled

Easier maintenance

